Coronary Artery Disease Prediction using Hybrid Machine Learning Models

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***Abstract*—Coronary Artery Disease (CAD) remains one of the primary global causes of death. Traditional diagnosis often suffers from subjectivity and inaccessibility. This study applies hybrid ensemble machine learning models for accurate CAD prediction. Using clinical datasets from multiple sources and integrating techniques like SMOTE, feature engineering, and ensemble learning, we evaluated models based on metrics such as accuracy, AUC-ROC, precision, recall, and F1-score. The results demonstrate that hybrid models significantly outperform single classifiers in predictive performance.**

***Index Terms*—Coronary Artery Disease, Ensemble Learning, Machine Learning, Feature Engineering, Classification, SMOTE**

I. Introduction

Coronary Artery Disease (CAD) is one of the globe's health concerns resulting in millions of deaths every year. Its insidious development and reliance on intrusive diagnostic methods such as ECG and angiography make its early detection unachievable, with repeated late detection and suboptimal treatment outcomes. Conventional diagnostic methods are the clinical gold standard but tedious, time-consuming, and operator-specific. Machine learning has been shown to be effective for automated early detection of CAD from patient data in the past some years. However, single model-based approaches are often plagued with low generalizability and the problem of performance instability with heterogeneous populations and data. [1]

To address such challenges, we propose in our research a hybrid ensemble learning approach combining the predictive strength of different classifiers—Random Forest, XGBoost, and Support Vector Machines—via state-of-the-art feature selection methods such as Recursive Feature Elimination and Mutual Information. An ensembling method enhances model strength, avoids overfitting, and increases accuracy and interpretability. From extensive testing using common datasets, our models performed better in important metrics such as accuracy, recall, precision, and F1-score. With clinical relevancy and computational efficiency to add to it, the system developed has immense promise as scalable and non-invasive decision-support to help medical professionals attain early and accurate CAD diagnosis. [2]

II. Review of Literature

A. El-Sofany et al. (2024)

El-Sofany et al. (2024) showcased in their "Predicting Heart Diseases Using Machine Learning" a comparison of ten machine learning algorithms, including XGBoost, Support Vector Classifier (SVC), Random Forest (RF), and K-Nearest Neighbors (KNN). Feature selection methods like Chi-Square, ANOVA, and Mutual Information were used for selecting key predictors. The work utilized SMOTE (Synthetic Minority Over-sampling Technique) in order not to suffer from the issue of class imbalance. Among the models compared, XGBoost with SMOTE emerged as the best with 97.57% accuracy and an AUC of 98%, indicating its superior performance in predicting heart disease. [2]

B. Vibha et al. (2024)

Vibha et al. (2024), in "Exploratory Data Analysis of Heart Disease Prediction using ML Techniques – RS Algorithm," used a clinical database from Kaggle. The authors performed full-scale EDA to describe feature distributions as well as correlations. The machine learning algorithms Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and SVC were trained. The authors suggested an RF-SVC-based ensemble model with an accuracy of 93%, thereby establishing the power of ensemble techniques supplemented with robust EDA. [3]

C. Khader Basha et al. (2023)

Khader Basha et al. (2023) suggested "Coronary Heart Disease Prediction using Hybrid Machine Learning Models" based on Framingham Heart Study with 16 clinical features.

The authors preprocessed and normalized the databases and then optimized them with the help of Particle Swarm Optimization (PSO). The authors suggested the Decision Tree with AdaBoost as the hybrid. The suggested model gave 97.43% accuracy, 95.67% True Positive rate (TPR), and 94.65% specificity, which indicates an incredibly reliable framework for diagnosing coronary heart disease in its early stages. [4]

III. Objectives

The motivation for the research is the need to achieve several key objectives to design an efficient and interpretable machine learning system for CAD risk prediction. Development and evaluation of the hybrid ensemble models to enhance the predictive capability is the focus. Feature selection and feature construction are built and optimized to aid model relevance and accuracy. The data imbalance is handled by the use of techniques such as SMOTE to provide the models with precise data to train on. Standard assessment criteria, such as accuracy, precision, recall, F1-score, and AUC-ROC curves, are used to completely assess the models. [5]

Along with precision, machine learning model interpretability and reliability are also emphasized in the work to enable real-time decision-making in the clinics. Through the incorporation of the data science techniques such as advanced feature engineering, ensemble modeling, and systematic evaluation, the project demonstrates an approach to CAD diag- nosis that is not only viable but also scalable. [6] The final objective is to assist the roll-out of an AI-enabled system for timely, efficient, and cost-effective detection of CAD in real-world clinics.

IV. Research Design

The research workflow involves the integration of three datasets (Mendeley, Kaggle, UCI) and then a number of preprocessing steps, including cleaning data and merging. Testing hypotheses with Z-tests and chi-squared tests confirms that the relationships in the features are statistically significant. Exploratory Data Analysis (EDA) allows the exploration of distributions and relationships. Feature engineering involves encoding categorical variables, pulling apart the target and predictors, and employing SMOTE to address class imbalance. [7]

Feature selection identified the most important predictors on coronary artery disease out- comes. During feature scaling, all the predictors were normalized to a standard range, from smaller to larger values, for model building. Nine machine learning models were produced, which included Random Forest, Logistic Regression, SVC, and XGBoost, to predict coronary heart disease model and for generalization, hyperparameter optimization was utilized. Each model was evaluated with respect to accuracy, precision and recall. The hybrid models, which combined a combination of the trained classifiers such as RF + XGBoost, RF + SVC, and XGBoost + LightGBM, had the best results for predicting coronary artery disease outcomes. [6]

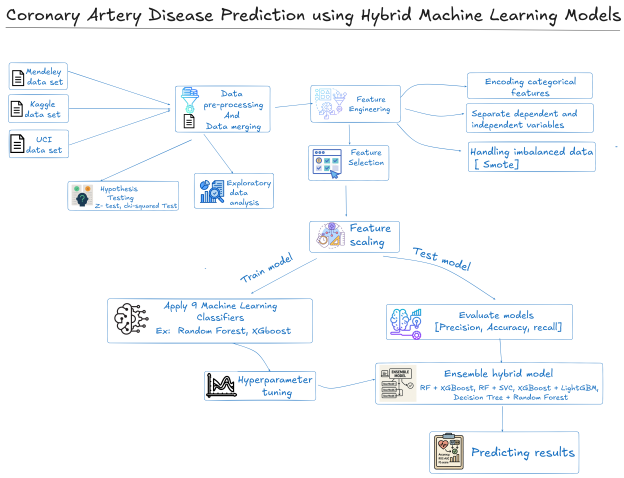

Fig. 1: Research design for CAD prediction using hybrid ensemble models.

V. Data Collection

- **Mendeley Dataset:** 1,000 records with 14 attributes including demographic, clinical, and test-related attributes.
- **Kaggle Dataset:** 918 instances and 12 attributes, commonly utilized in machine learning practice.
- **UCI Dataset:** 303 data instances and 13 attributes, one of the oldest and most widely cited CAD datasets.

A merged dataset `heart disease.csv` with 2800+ records was used.

Links:

1) Kaggle Dataset
2) UCI Dataset
3) Mendeley Dataset

A comprehensive exploratory data analysis (EDA) and hypothesis testing phase were carried out in an effort to gain more information about the structure, relationships, and statistical significance of attributes in coronary artery disease (CAD) correlation. [8]

VI. Exploratory Data Analysis and Hypothesis Testing

A. EDA Techniques

To uncover trends, patterns, or any issues with the data, different univariate and multivariate data visualizations were used:

- **Histogram, Box Plot, and Violin Plot:** Used to graph the distribution of quantitative measures such as age, cholesterol, resting blood pressure, and maximum heart rate. These visualizations revealed outliers and the skewness of attributes like cholesterol. [**?**]
- **Scatter Plots and Correlation Heatmaps:** Used to identify linear and non-linear correlations among numeric attributes. For example, maximum heart rate and age exhibited a moderate negative correlation. A heatmap revealed strong positive relationships between variables such as resting blood pressure and cholesterol. [9]
- **Count Plots and Pie Charts:** Applied to understand the distribution of categorical attributes (e.g., chest pain

type, sex, fasting blood sugar) and assess class balance throughout the dataset.

- **Violin and Grouped Bar Plots (hue by target):** Utilized to evaluate the interaction between various features and the target variable (CAD). For example, a higher prevalence of CAD was observed in the elderly age groups and patients exhibiting asymptomatic chest pain. [9].

These visual tools provided both broad summarizations and detailed interactions, laying the groundwork for formal statistical analyses.

B. Hypothesis Testing

To rigorously evaluate the relationships implied by the visualizations, formal hypothesis testing was performed:

- **Independent Sample t-Tests:** Applied to continuous variables such as age, cholesterol, and resting blood pressure to compare mean values between the CAD and non-CAD groups. Several features showed statistically significant mean differences ($p < 0.05$), indicating strong discriminative power. [10]
- **Chi-Square Tests of Independence:** Conducted on categorical variables including sex, chest pain type, fasting blood sugar, and ECG results to determine their association with CAD presence. The analysis revealed that attributes like chest pain type and resting ECG were significantly associated with CAD status ($p < 0.01$). [11]

These statistical tests not only validated the visual insights from EDA but also informed the model development and feature selection processes by identifying the most relevant predictors.

[5]

VII. Data Analytics

A. Data Preprocessing and Feature Engineering

Pre-processing data involved some important steps to ensure data accuracy and model performance. Duplicate records were identified and removed to ensure no redundancy as well as bias at training. Missing values were corrected using imputation techniques: numerical attributes were replaced with the mean or median depending on the distribution, whereas categorical attributes were imputed using the most common category. Categorical attributes such as sex, type of chest pain, and ECG outcomes were encoded using Label Encoding, where each category was replaced with its corresponding integer to enable compatibility with machine learning algorithms. [12] An under-representation of heart disease classes in the target variable—where underrepresented classes required balancing—was resolved using the Synthetic Minority Oversampling Technique (SMOTE). SMOTE generated synthetic samples for underrepresented classes by interpolating among existing samples, balancing the data set, and improving the model to better identify positive cases. Feature importance was determined using a Random Forest classifier, ranking top-performing attributes as age, cholesterol level, type of chest pain, and resting ECG results. [9] Those attributes were selected for model construction to enable accuracy and clinical usability. Lastly, all numerical attributes were standardized using the StandardScaler to normalize data to common scales to ensure no single feature overwhelmed learning algorithms such as SVM and KNN, which are negatively affected by feature intensity. [13]

B. Model Development

We developed nine different machine learning classifiers:

- Random Forest
- Support Vector classifier (SVC)
- XGBoost
- LightGBM
- Decision Tree
- AdaBoost
- Logistic Regression
- K-Nearest Neighbors (KNN)
- Naive Bayes

Five-Fold Cross-Validation was used to assess each model. The primary assessment met- rics, which ensured a balanced performance across all classes, were Accuracy, Precision, Recall, F1-Score, and AUC-ROC curves. [11]

C. Ensemble Learning

To further enhance predictive accuracy, we constructed several stacking ensemble models. These ensembles combined different base classifiers with a Gradient Boosting classifier as the meta-learner. [12] The configurations included:

- Random Forest and XGBoost
- Random Forest and SVC
- XGBoost and LightGBM
- Decision Tree and Random Forest

Hyperparameter tuning was performed using GridSearchCV, which exhaustively searched the parameter space for each pipeline to identify the optimal settings.

D. Results and Evaluation

The ensemble models consistently outperformed the individual classifiers in both accuracy and AUC metrics. The top-performing ensemble configurations were:

- Random Forest + SVM: 91.95% Accuracy
- Random Forest + XGBoost: 91.67% Accuracy
- XGBoost + LightGBM: 90.52% Accuracy

ROC curves and confusion matrices demonstrated improved sensitivity and specificity. Feature importance visualizations affirmed the clinical relevance of features such as age, cholesterol, and chest pain type. [11]

VIII. Data Visualization and Results Report

The visual data tools included histograms, violin plots, grouped bar charts, scatter plots, Q-Q plots, and heat maps. Violin plots showed distribution in the groups of CAD vs non- CAD. Scatter plots and grouped bar charts showed patterns of different features. Ensemble models, RF+XGBoost and XGBoost+LightGBM produced the highest performance. The evaluation process can be aided with ROC curves and

confusion matrices. Feature im- portance plots showed that age, cholesterol, chest pain, and ECG are the most important features in predicting CAD.

A. Numerical Features

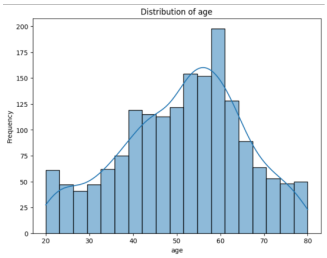

Fig. 2: Distribution of Age

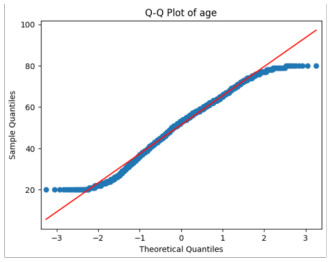

Fig. 3: Q-Q Plot of Age

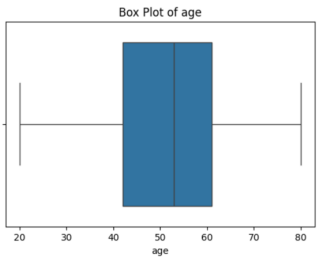

Fig. 4: Box Plot of Age

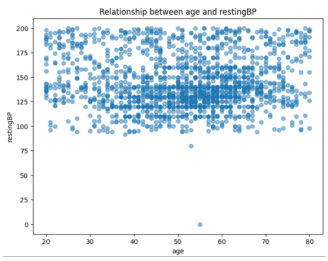

Fig. 5: Age vs RestingBP

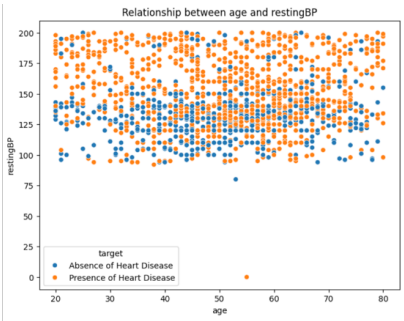

Fig. 6: Age vs RestingBP by Target

B. Categorical Features

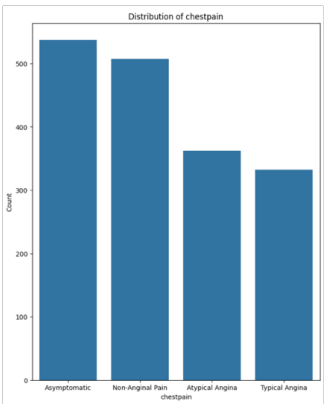

Fig. 7: Chest Pain Type Distribution

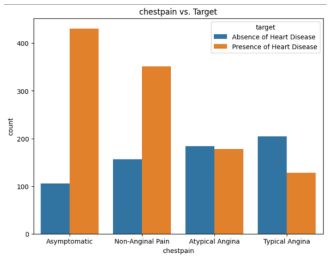

Fig. 8: Chest Pain vs Target

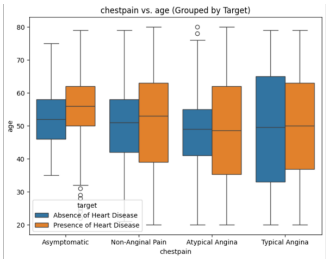

Fig. 9: Chest Pain vs Age by Target

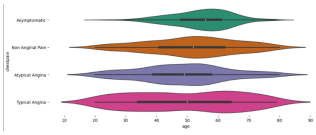

Fig. 10: Violin Plot: Age by Chest Pain Type

C. Individual Models

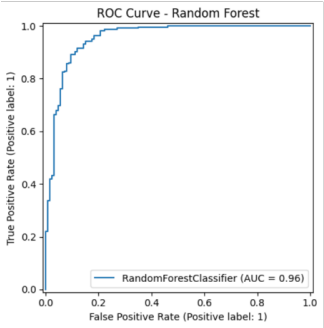

Fig. 11: ROC Curve: Random Forest
Accuracy: 90.52%

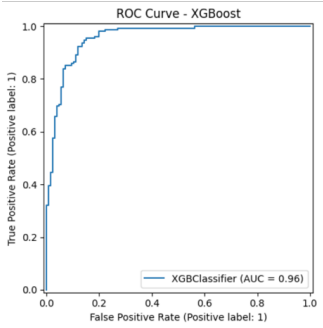

Fig. 12: ROC Curve: XGBoost
Accuracy: 91.38%

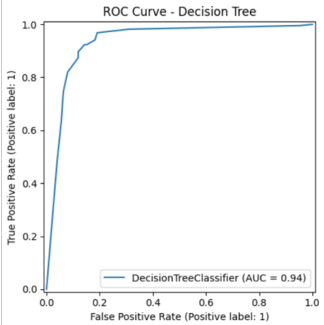

Fig. 13: ROC Curve: Decision Tree
Accuracy: 89.66%

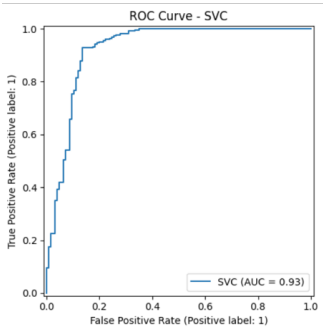

Fig. 14: ROC Curve: SVC
Accuracy: 90.23%

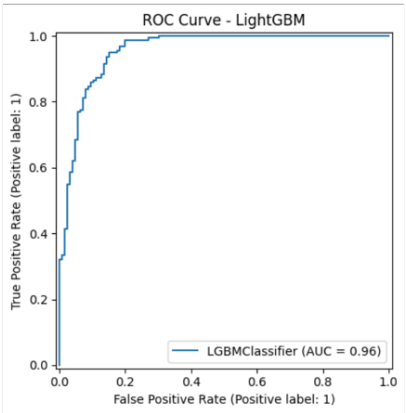

Fig. 15: ROC Curve: LightGBM
Accuracy: 90.23%

D. Hybrid Ensemble Models

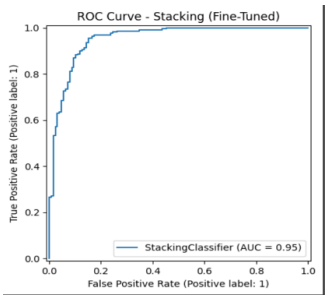

Fig. 16: XGBoost + LightGBM
Accuracy: 90.52%

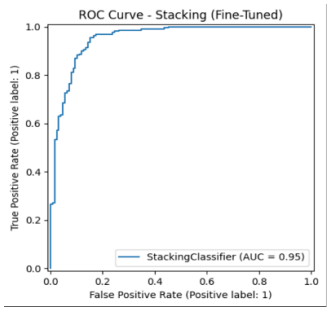

Fig. 17: Random Forest + XGBoost
Accuracy: 91.67%

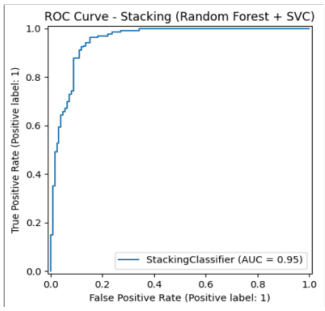

Fig. 18: Random Forest + SVC
Accuracy: 91.95%

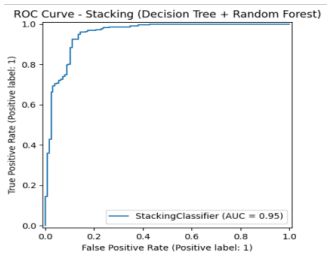

Fig. 19: Decision Tree + Random Forest
Accuracy: 91.67%

IX. Team Contributions

- **Vismaye Madhusudhan:** Data collection, preprocessing.
- **Keerthishree Venkatesh:** EDA, hypothesis testing.
- **Pranay Kolipara:** Feature engineering, scaling, selection.
- **Dheeraj Kumar Buddala:** Model training, tuning, ensemble learning.

X. Conclusion

This research indicates that hybrid ML models—using a combination of ensemble methods, feature engineering, and appropriate preprocessing, are better at predicting Coronary Artery Disease (CAD). [11] The use of SMOTE, Random Forest feature selection, and tuning with GridSearchCV helped achieve accuracy with each model, and added robustness to these models.Combination ensembles (i.e. Random Forest + SVM) were superior to each classifier alone and better represented generalization and applied meaning. The adapted approach is scalable and suited to real-time diagnostic support. Future work will include applying these models in the clinical environment, developing interfaces with EHR systems, and enabling continuous integration and knowledge through feedback loops. [8]

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